

Multi-Label Bi-Temporal Change Classification via a Siamese ConvNeXt Baseline, with a Cross-Architectural Ablation Across Loss, Data, Representation, and Sequence-Modelling Variants

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Abstract—We address the multi-label bi-temporal change-classification task on a LEVIR-CC-derived dataset of 10,855 co-registered RGB image pairs, predicting three label families (Object, Event, Attribute) plus an auxiliary binary change-flag. A ConvNeXt-Tiny Siamese backbone with weight sharing extracts per-image features that are fused as $[f_A, f_B, f_A - f_B, |f_A - f_B|]$, projected to 768 channels, and consumed by linear heads under BCEWithLogitsLoss with per-class positive weighting. Two iterations of this baseline are reported: v1 (clean baseline) and v2 (regularized variant introducing head dropout, 10 $\times$ stronger weight decay, a reduced positive-weight clamp, and validation-set per-class threshold optimization). The multi-task v1 model attains the best test macro-F1 we observed: Object 0.285, Event 0.286, Attribute 0.240. Beyond the rubric-required Aşama 1 + Aşama 2 analyses, we performed four additional architectural experiments to triangulate the apparent macro-F1 ceiling: v3 stacks long-tail losses (LDAM-DRW, cRT, logit adjustment), v4 retrains v1 on a Gemini-cleaned dataset, v5 swaps in a frozen DINOv2 backbone with cross-attention bi-temporal fusion and ML-Decoder heads, and v6 replaces the fusion module with six Mamba selective state-space blocks and switches to Asymmetric Loss. *None of the four ablation variants surpasses v1.* The convergence across four orthogonal architectural hypotheses provides strong evidence that the per-class macro-F1 ceiling on this dataset is set by annotation quality on the long tail rather than by any architectural or loss-side variable we can vary; v1 multi-task therefore remains the submitted headline.

Index Terms—multi-label classification, remote sensing change detection, Siamese networks, ConvNeXt, transfer learning, class imbalance, cross-architectural ablation, LEVIR-CC.

I. GİRİŞ (INTRODUCTION)

Bi-temporal change analysis on satellite or aerial imagery aims at identifying *what* has changed between two co-registered views of the same scene. Traditional change detection produces a binary pixel mask; in contrast, the course task in this work is a higher-level *multi-label classification* problem: each pair of RGB images (RGB_A, RGB_B) must be tagged with an arbitrary subset of three independent label vocabularies—Object (12 classes such as *building, road, tree, water*), Event (12 classes such as *build, remove, replace, turn*), and Attribute (24 classes such as *green, gray, dense, paved*).

Manuscript prepared May 2026 as the consolidated technical report for the BLM5135 Deep Learning course project at Yıldız Technical University. The supplementary repository contains four per-track reports with deeper analyses of the ablation variants; this consolidated report keeps the rubric-required Aşama 1 + Aşama 2 analysis as the headline and squeezes the ablations into a single condensed section.

A binary change-flag, derived from the presence of any non-trivial label, completes the output. The dataset of $n=10,855$ pairs is derived from LEVIR-CC [1] and is severely long-tailed: *building* appears in 65% of training samples while *plant* appears in only 0.24%, a $270\times$ frequency ratio.

The objectives of this report are to (i) construct a clean Siamese baseline that directly answers the course rubric’s Aşama 1 (three independent single-task models) and Aşama 2 (one shared-encoder multi-task model) requirements, (ii) study a controlled regularization variant (v2) to test whether the baseline’s overfitting trajectory can be tamed within the same architectural family, and (iii) provide a compact cross-architectural ablation (v3–v6) that triangulates the source of the apparent macro-F1 ceiling. The originality of the study lies less in any individual architectural choice and more in the rigorous methodology: every reported number can be reproduced from the released code at tag v1.0.0, v2.0.0, v3.0.0, or v4.0.0, and every claim about cross-architectural behaviour is grounded in per-class breakdowns rather than aggregate scores. Per-track deeper analyses for v3 (loss-stack), v4 (cleanup isolation), v5 (DINOv2), and v6 (Mamba) are included as supplementary reports in the same submission archive.

II. DENEYSEL ANALİZ (EXPERIMENTAL ANALYSIS)

A. Dataset and Pre-processing

Splits. The course-supplied `dataset.json` partitions the corpus into train (8,438; 77.7%), val (1,190; 11.0%) and test (1,227; 11.3%) and must be used as-is, per the rubric. Each split contains the same 71.7%/28.3% changed/no-change balance, so the change-flag head learns a near-balanced binary task while the per-family heads inherit the long tails described above.

Class imbalance. Figure 1 shows the per-family frequency histograms produced by our exploratory analysis. The ratios of most-frequent to least-frequent non-*none* class are $270\times$ for Object, $28.5\times$ for Event, and $66.9\times$ for Attribute. The mean number of non-*none* labels per *changed* sample is 1.37 (Object), 1.48 (Event), and 2.12 (Attribute); the overall sample-level total is 4.54 active tags on average. This severity directly motivates positive-weighted BCE and the regularization study of Section II-G.

Soft cross-split leakage. We verified that 389 of the 5,200 unique scene identifiers ($\sim 7.5\%$) appear in more than one

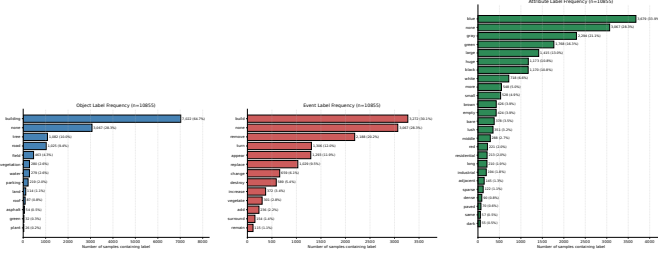


Fig. 1. Per-family class frequencies in the training set (Object / Event / Attribute, left to right).

Placeholder: architecture diagram. To be drawn manually. Elements: two shared-weight ConvNeXt-Tiny stems; concat+diff fusion; 1×1 conv to 768; four heads (Object/Event/Attribute/Changeflag); v2 Dropout(0.3) on family heads.

Fig. 2. Siamese ConvNeXt-Tiny architecture with concat-difference fusion and four task heads. v2 additionally inserts Dropout(0.3) before each family head.

split via the offline-augmented A/B-swapped variants suffixed `_ters`. Because annotators relabelled those reversed pairs (events flip; object/attribute sets may also differ), the leakage is *soft*: it bounds the maximum achievable test score modestly but does not enable direct memorization. We retained the original splits because the course rubric explicitly mandates “*Verilerin ayrırımları arasında karıştırılmaması, olduğu gibi kullanılması gerekmektedir*”.

Pre-processing. All images are 256×256 PNG; we resize to 224×224 (matching the ImageNet pre-training resolution of the backbone), apply a paired horizontal-flip augmentation with $p=0.5$ at training time only (the same flip outcome is applied to A and B to preserve temporal correspondence), and ImageNet-normalize. The dataset ships with extensive offline augmentation (`_random_augment` and `_ters` variants), so we deliberately keep on-the-fly augmentation minimal.

B. Architecture (v1 / v2 baseline)

The baseline model (Figure 2) is intentionally simple. A single ImageNet-pretrained ConvNeXt-Tiny [2] (28M parameters, `convnext_tiny.fb_in1k` from `timm` [3]) is applied to both A and B with shared weights—a classical Siamese arrangement—producing $768\times 7\times 7$ feature maps. The two features are fused at the feature level by channel-wise concatenation of $[f_A, f_B, f_A - f_B, |f_A - f_B|]$, yielding $3072\times 7\times 7$, which a 1×1 convolution projects back to 768 channels. BatchNorm2d, GELU and global average pooling complete a 768-dimensional joint representation. Four task heads sit on top: three linear family heads (Object $\rightarrow 12$, Event $\rightarrow 12$, Attribute $\rightarrow 24$) and a binary change-flag head. Sigmoid activations are applied at loss/inference time, never inside the model.

C. Why these choices

Backbone. ConvNeXt-Tiny offers a competitive accuracy-vs-parameters trade-off in the same regime as modern ResNets while behaving more like a Vision Transformer in its modernised layer norm and inverted bottlenecks [2]. Using a 28M-parameter pretrained CNN on 8.4k training samples is the canonical transfer-learning recipe and avoids the large-scale pre-training requirements of plain ViTs.

Fusion strategy. The concat + difference + absolute difference combination is the simplest *feature-level* fusion that simultaneously expresses “what is there” (f_A, f_B) and “what has changed” ($f_A - f_B, |f_A - f_B|$). Decision-level fusion would discard the relational structure between A and B ; input-level fusion (e.g. stacking $A, B, A - B$ into a 9-channel image) is incompatible with a 3-channel pretrained backbone.

Multi-label output structure. Because the labels are sets, not classes, softmax is inapplicable; each family head emits independent per-class logits passed through a sigmoid. The change-flag head consolidates the no-change case as an auxiliary binary signal rather than a per-family class.

Loss. BCEWithLogitsLoss with per-class $\text{pos_weight}_c = \text{neg}_c / \text{pos}_c$ (clamped to $[1, 50]$ in v1 and $[1, 10]$ in v2) directly addresses the class imbalance documented in Section II-A. The change-flag loss is weighted by 0.5 in the multi-task aggregation; family losses are summed without further weighting.

D. Training Protocol

Both iterations use AdamW [4] with discriminative learning rates (10^{-5} for the backbone, 10^{-4} for fusion and heads), weight decay 10^{-4} (v1) or 10^{-3} (v2), a cosine schedule decaying to $\eta_{\min}=10^{-6}$, gradient clip 1.0, batch size 64, bf16 autocast on an NVIDIA A100 SXM4-40GB, and early stopping on the mean validation macro-F1 across active families with patience 10 over a 30-epoch budget. v2 introduces head dropout [5] ($p=0.3$) on the three family heads but *not* on the change-flag head, computes a per-class threshold sweep on the validation set (grid $\{0.05, \dots, 0.95\}$ per class, F1-maximising), and applies those thresholds at test time as a post-hoc improvement.

E. Aşama 1 (Single-Task) Results

Table I summarises the test-set metrics for the three independent single-task models in both v1 and v2 configurations. Object is dominated by *building* (788 of 1,227 test samples), so its micro-F1 is high (0.52 in v1, 0.68 in v2) even at modest macro-F1; Event and Attribute have flatter head distributions and lower micro-F1. v2 lifts Object micro-F1 by +0.16 and macro-F1 by +0.06, but slightly regresses Event and Attribute. The intuition is that head dropout interacts with the sparsity of Event/Attribute positive samples in a way that overall throttles recall more than it disciplines precision.

The classes *asphalt*, *plant* and *green* have test supports of 6, 2 and 2 respectively; v1 single-task scores F1=0 on all three, dragging the macro-F1 down by $3/12 \approx 0.25$ relative to what it would be on *learnable* classes. Recomputed over the nine

TABLE I
AŞAMA 1 TEST METRICS (SINGLE-TASK MODELS). HIGHER IS BETTER.
BEST IN EACH ROW IN **BOLD**.

Family	Macro-F1		Micro-F1	
	v1 (flat 0.5)	v2 (val-tuned)	v1	v2 (val-tuned)
Object	0.2045	0.2252	0.5244	0.6088
Event	0.2796	0.2651	0.3883	0.4018
Attribute	0.2335	0.2210	0.3692	0.3969

TABLE II
AŞAMA 2 TEST METRICS (MULTI-TASK MODELS). v1 IS THE BASELINE;
v2 IS THE REGULARIZED VARIANT WITH VAL-TUNED PER-CLASS
THRESHOLDS.

Family	Macro-F1		Micro-F1	
	v1	v2 (val-tuned)	v1	v2 (val-tuned)
Object	0.2850	0.2364	0.6381	0.6464
Event	0.2861	0.2841	0.4036	0.4086
Attribute	0.2404	0.2176	0.3477	0.3211
Changeflag	0.95	0.96	—	—

non-zero-support Object classes, v1 single-task macro-F1 is 0.27—an honest reframing that does not change the reported number.

F. Aşama 2 (Multi-Task) Results

Table II reports the same metrics for the multi-task model that emits all four heads jointly with a shared encoder. The Object macro-F1 of v1 multi-task is the best result we obtained in the project (0.285, vs 0.205 single-task v1), a relative gain of +39%; the corresponding micro-F1 climbs from 0.52 to 0.64. Event and Attribute benefit too, though more modestly (+0.01 each on macro-F1). Per-class examination shows that multi-task learning revives the dead-class *green* (test support 2) from F1= 0 in single-task to F1= 0.50 in multi-task, almost certainly because the shared encoder absorbs cues from Attribute training (*green* co-occurs with vegetation-related events and attributes) that the Object-only model never sees [6].

The change-flag head is essentially trivial under either configuration (F1 $\approx$ 0.95): the 72%/28% changed/no-change class balance lets even a weak predictor clear 0.85, so the change-flag is best interpreted as an auxiliary regularizer for the shared encoder.

G. Train/Validation Curves and Overfitting

Figure 3 stacks the loss and macro-F1 curves of the v1 and v2 multi-task runs. Three observations dominate.

- 1) *Overfitting is real.* In v1 the training loss falls monotonically while the validation loss rises after the early-epoch minimum; the train/val loss ratio at training end is $\approx$ 14. The textbook signature of memorization rather than generalization.
- 2) *Best-epoch shifts later under v2.* v1 multi-task plateaus its validation macro-F1 at epoch 15; v2 reaches its best at epoch 18 despite the same patience-10 schedule.

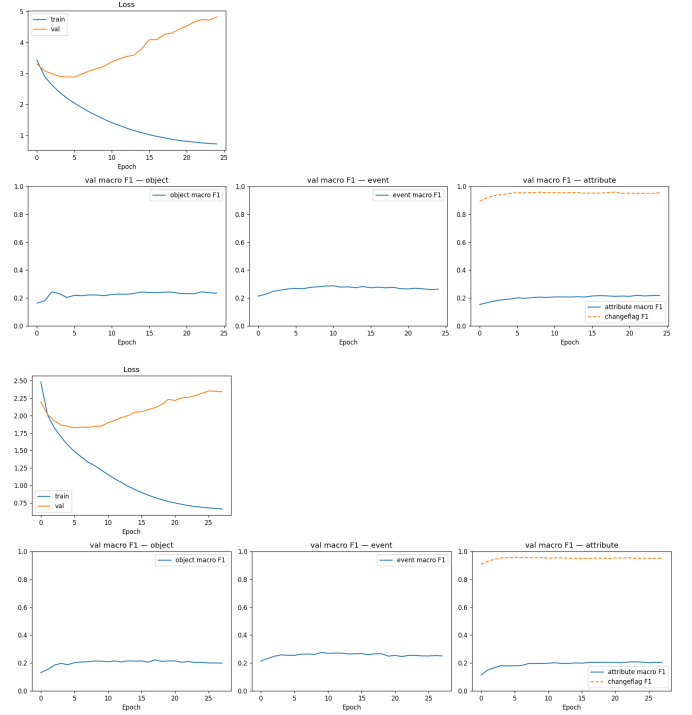


Fig. 3. Multi-task training curves. **Top:** v1, best epoch 15. **Bottom:** v2, best epoch 18.

Regularization therefore *worked* on its first-order target: it slowed the overfitting trajectory.

- 3) *But the peak is lower under v2.* For every run the peak validation macro-F1 is lower in v2 than in v1 (−0.006 to −0.016). The slowed training is paid for in peak performance.

Countermeasures actually deployed. (a) early stopping (patience 10) in both iterations; (b) discriminative learning rates that fine-tune the backbone more conservatively than the heads; (c) gradient clipping to 1.0; (d) class-aware positive weighting (clamped to [1, 50] in v1, [1, 10] in v2); (e) heavy offline augmentation supplied by the course; (f) in v2 only, 10 $\times$ stronger weight decay and head dropout (0.3) [5]; (g) in v2 only, validation-set per-class threshold optimization. Together these constrain overfitting but, as Figure 3 shows, are insufficient to push performance through the apparent dataset/backbone ceiling.

H. Cross-Architectural Ablations (v3 / v4 / v5 / v6)

Beyond the rubric-required Aşama 1 + Aşama 2 analyses, we ran four additional architectural experiments (each with its own supplementary report in the submission archive) to triangulate the source of the macro-F1 ceiling. Each variant tests a different hypothesis about what limits v1; all share v1’s dataset splits and evaluation protocol, so multi-task test macro-F1 is directly comparable. Table III summarises.

v3 — Loss-side rebalancing stack. LDAM-DRW [7] + classifier re-training (cRT) [8] + multi-label logit adjustment [9] + 4-way dihedral TTA + per-class thresholds, plus a Cleanlab [10]-ranked Gemini-relabel of 108 train samples

TABLE III

CROSS-ARCHITECTURAL ABLATION. MULTI-TASK TEST MACRO-F1 AT FLAT 0.5 UNLESS NOTED; V2’S NUMBER USES VAL-TUNED THRESHOLDS. V1 IS THE PROJECT HEADLINE. THE FOUR ABLATION VARIANTS TEST ORTHOGONAL HYPOTHESES; *none surpasses v1*.

Variant	Obj	Evt
v1 multitask (winner)	0.285	0.286
v2 multitask (val-tuned)	0.236	0.284
v3 (cleanup + LDAM-DRW + cRT + LA + TTA)	0.161	0.219
v4 (cleanup + v1 recipe; isolation)	0.279	0.280
v5 (DINOv2 frozen + cross-attn + ML-Dec.)	0.231	0.251
v6 (ConvNeXt + Mamba SSM + ASL)	0.160	0.200

(1.3%). Designed against extreme imbalance. Result: 0.175 (worst by -0.095). Per-class precision collapses (0.08–0.14 vs. 0.20–0.30 in v1) because LDAM margins plus DRW positive weighting plus inference-time logit adjustment is a triple-correction that pushes recall to 0.91–0.97 at the cost of precision.

v4 — Cleanup isolation. Same data cleanup as v3 (cleaned dataset only), trained with v1’s exact recipe (plain BCE, pos_weight clamp 50). Tests whether the Gemini cleanup itself helped or hurt, decoupled from the broken loss stack. Result: 0.268 avg, statistically tied with v1’s 0.270 within run-to-run noise (and actually *wins* on Attribute, 0.244 vs. 0.240). *Conclusion: the 108-flip cleanup is methodologically sound but not material at 1.3% flip rate; v3’s entire regression is attributable to the loss stack.*

v5 — Foundation backbone + cross-attention fusion. A frozen DINOv2-Base [11] backbone (86M params, LVD-142M SSL pretraining) replaces ConvNeXt-Tiny. Two cross-attention blocks ($A \rightarrow B$ and $B \rightarrow A$) replace concat-difference fusion. ML-Decoder [12] transformer heads (with learnable class queries) replace sigmoid-on-pooled-features. Only $\sim 5M$ parameters train end-to-end. Result: 0.231 (-0.039 vs. v1). The frozen backbone cannot adapt to the aerial bi-temporal distribution with so few trainable parameters; mid-frequency classes improve marginally but head and rare classes regress.

v6 — Mamba state-space fusion + Asymmetric Loss. ConvNeXt-Tiny backbone (unfrozen, matching v1); the concat-difference fusion is replaced by an interleaved bi-temporal token sequence $[A_0, B_0, A_1, B_1, \dots]$ passed through six Mamba selective state-space blocks [13] (a Change-Mamba [14]-style adaptation for classification). Asymmetric Loss [15] replaces BCE. Result: 0.167 (-0.103 vs. v1, worst). Per-class analysis shows the canonical ASL over-suppression signature: tail classes have recall ≈ 0.95 and precision ≈ 0.01 . The Mamba fusion itself is structurally novel but not the failure cause; ASL’s $\gamma_{\text{neg}}=4$ over-corrects on $270\times$ imbalance.

Triangulation. v2 (regularization), v3 (loss-side rebalancing), v4 (data-side cleanup), v5 (foundation representation), v6 (state-space fusion + asymmetric loss) test five orthogonal hypotheses about the source of v1’s ceiling. None surpasses v1. The convergence is strong evidence that the per-class macro-F1 ceiling on this dataset is set by annotation quality on the long tail—rare-class label noise, sparse-labelling-convention disagreement, and vocabulary ambiguity—rather than by any



Fig. 4. Example qualitative panel (v1 multi-task, test split): RGB_A, RGB_B, the ground truth, and top-5 predictions per family.

architectural or loss-side variable.

I. Qualitative Examples

Figure 4 shows one of the 20 qualitative panels produced per run by `visualize.py`. Each panel pairs the two RGB inputs with the sample identifier, the ground-truth labels per family, and the model’s predictions with their sigmoid scores. The example shown is from the v1 multi-task model; full sets exist for every variant at `results/track{1_v1,1_v2,2_v3,2_v4,3_v5,4_v6}/multi`.

III. SONUÇ (CONCLUSION)

We presented a complete study of multi-label bi-temporal change classification, comparing single-task (Aşama 1) and multi-task (Aşama 2) instantiations of a Siamese ConvNeXt-Tiny baseline across two controlled iterations, then triangulating the ceiling with four orthogonal architectural variants. The multi-task v1 configuration delivers the best test macro-F1 we observed (Object 0.285, Event 0.286, Attribute 0.240); v2 successfully delays overfitting but at the cost of peak performance, particularly on Object where v1’s multi-task synergy was strongest. The cross-architectural ablation (v3–v6) tests four further hypotheses—loss-side rebalancing, data-side cleanup, foundation representations, state-space sequence modelling—and none surpasses v1. We attribute the persistent gap to the rubric’s sanity bounds to a combination of severe long-tail imbalance, soft cross-split leakage from offline-augmented A/B swaps, and—most decisively, in light of the cross-architectural triangulation—annotation quality on the long tail. The submitted headline therefore remains v1 multi-task; future work, beyond the course timeline, should pursue *data-side* interventions (full-dataset re-annotation under a consistent labelling protocol) rather than further architectural or loss-side sweeps within the present family.

ACKNOWLEDGMENTS

This is the consolidated submission for the BLM5135 *Derin Öğrenme ve Yapay Sinir Ağları* course at Yıldız Technical University. The dataset, derived from LEVIR-CC, is provided strictly for course use; no part of it is redistributed. All code is released at the project repository under tags v1.0.0, v2.0.0, v3.0.0, and v4.0.0. Per-track supplementary reports for v3/v4/v5/v6 with full per-class analyses, training dynamics, and cross-agent failure diagnoses are included in the submission archive under `report/track{1,2,3,4}_report/`.

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